F28-SR-002
ULTRA FINE GRINDING 415V SWITCHROOM

F28-SW-009
FINAL TAILS PUMPS 415V MCC
2000-F28-TER-IC-10101/10102

(NOTE 4)
F28-ETH-021
F28-ETH-025
INCOMER
F28SW009-GWY
F28SW009-INC INCOMER ACB
F28SW009-PQM POWER METER
F28SW009-AFD ARC FLT. DET. SYS.
RS485 BUS
EOL 1
EOL n
DOL STARTER MODULES ON FRONT & REAR OF MCC
(NOTE 1, 2 & 3)

LV VSDS
1
n
F28-UPS-002
240VAC UPS
(NOTE 6)
F28-FIP-001
FIRE PANEL

F28-SR-001
ULTRA FINE GRINDING 11KV SWITCHROOM

F28-SW-013
ULTRAFINE GRINDING
11KV SWITCHBOARD
2000-F28-TER-IC-10001
F28SW013-PQM POWER METER
PANEL 1 RELAY F28SW013-IED01
PANEL n RELAY F28SW013-IEDnn
(NOTE 1, 2 & 3)

F28-MUF-402
ULTRA FINE GRINDING MILL 2
2000-F28-SCM-EL-10103
MOTOR RTD RELAY JUNCTION BOX
F28MUF402-JB14

F28-MUF-403
ULTRA FINE GRINDING MILL 3
2000-F28-SCM-EL-10107
MOTOR RTD RELAY JUNCTION BOX
F28MUF403-JB14

MULTI MODE FIBRE OPTIC CABLE
MULTI MODE FIBRE OPTIC CABLE

F28-COM-001
ULTRA FINE GRINDING
415V SWITCHROOM COMMUNICATIONS RACK
2000-F28-TER-IC-10001
F28-ETH-001 (MODBUS COMMS SWITCH 1)
F28-ETH-002 (MODBUS COMMS SWITCH 2)
STACK CABLES NOTE 1.
SWITCHROOM MODBUS CONNECTIONS DISTRIBUTED ACROSS 2 ETHERNET SWITCHES
F28-ETH-011 (VNET IP1 LAN DCS SWITCH)
F28-ETH-012 (VNET IP2 LAN DCS SWITCH)
F28-ETH-003 (CCTV)
F28-FOB-001
FIBRE BACKBONE THROUGH-PATCHING AS REQ'D
F28-FOB-002
F28-FOB-004A F28-FOB-004B
F28-FOB-005A F28-FOB-005B
F28-FOB-006A F28-FOB-006B
(NOTE 1, 2 & 3)

F28-DCS-410
ULTRA FINE GRINDING 415V SWITCHROOM DCS PANEL
2000-F28-TER-IC-10011
F28-FCS-410
F28FCS410-NU01
(NOTE 1, 2 & 3)

F28-GE-109
ULTRAFINE GRINDING ESSENTIAL SUPPLY 415V GENERATOR
2000-EL019-D04-10001

TO F13-SR-003
SAG & BALL MILL VSD SWITCHROOM
2000-F13-BLK-IC-10021

TO F72-SR-001
REAGENTS
415V SWITCHROOM
2000-F72-BLK-IC-10001

REAGENTS RIO PANEL RING
(F72-RIO-002)
2000-F28-BLK-IC-10002

ULTRA FINE GRINDING RIO PANEL RING 1
(F28-RIO-001 TO F28-RIO-002)
2000-F28-BLK-IC-10002

ULTRA FINE GRINDING RIO PANEL RING 2
(F28-RIO-003 TO F28-RIO-004)
2000-F28-BLK-IC-10002

NOTES

1. FOR INTERNAL FIBRE OPTIC, CAT6 PATCH LEAD (& STACK CABLING IN SWITCHROOM COMMS RACKS) EQUIPMENT CONNECTION DETAILS REFER TO SPECIFIC TERMINATION DIAGRAMS.
2. REFER TO SPECIFIC COMMUNICATIONS TERMINATION DRAWINGS FOR EXTERNAL EQUIPMENT CONNECTION DETAIL.
3. COM RACK, DCS PANEL, RIO PANEL, MCC & SWITCHBOARD HARDWARE/INTERCONNECTIONS (E.G. NIO NODE) & INTER-FOBOT PATCHING NOT SHOWN FOR CLARITY. REFER TO SPECIFIC COMMUNICATIONS TERMINATION DRGS FOR DETAIL.
4. ETHERNET SWITCHES LOCATED IN CABLE ZONE OF 415VAC MCC, FRONT & BACK - AS REQUIRED.
5. REMOTE CCTV CAMERA REQUIRES PRE-TERMINATED FIBRE OPTIC CABLE WITH MEDIA CONVERSION EQUIPMENT AT THE RIO PANEL AND REMOTE CCTV CAMERA JUNCTION BOX.
6. FIRE INDICATION PANELS SHALL BE ON A FIBRE OPTIC RING AND WILL TIE-IN TO EXISTING MAIN PLANT FIRE ANNUNCIATOR PANEL. FIP CONNECTS TO FIBRE INFRASTRUCTURE VIA 2 FIBRE OPTIC PATCH CABLES.

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

LEGEND

- FOB — FIBRE OPTIC BREAK OUT TRAY
- ETH — ETHERNET SWITCH HIRSCHMANN = DCS NETWORKS CISCO = ALL OTHER NETWORKS
- FCS — YOKOGAWA FIELD CONTROL UNIT
- xxxFCS4xx-NUxx — YOKOGAWA NODE INTERFACE UNIT (DUAL OPTICAL ESB PORTS: U = UPLINK, D = DOWNLINK) TAG RELATED TO PARENT FCS
- GWY — COMMUNICATIONS GATEWAY
- MC — FIBRE TO CAT6 MEDIA CONVERTER
- xxx — EQUIPMENT OR VENDOR PACKAGE PANELS AS NOTED WITHIN BLOCK.
- EOL — MCC MODULE DOL INTELLIGENT ELECTRONIC DEVICE/OVERLOAD
- UPS — UNINTERRUPTIBLE POWER SUPPLY
- VSD - VARIABLE SPEED DRIVE (WALL MOUNTED)
- MULTI CORE FIBRE OPTIC CABLE OR DUPLEX PATCH LEADS
- CAT6 — CAT6 ETHERNET CABLE CONNECTION

2000-F28-BLK-IC-10001_Rev0

F28-SR-001/F28-SR-002 switchrooms & N-IO panels
- communications block diagram